

Roll No.

Total No. of Questions : 06]

[Total No. of Printed Pages : 05

ESS-188

B.Tech. (New Scheme) Examination, 2025

(Fourth Semester)

MICROPROCESSOR AND

MICROCONTROLLER

EL-405

Time : 3 Hours]

[Maximum Marks : 60

Note : Q. No. 1 is compulsory. Attempt any *two* parts from remaining questions.

1. Write short answers type questions : **5×2=10**

(a) If the last address of 8K byte ROM is 7FFFH.

What will be the first address ?

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(b) What is the function of Direction and Overflow flag of 8086 ?

(c) What are the advance features available in IC8254 compared to IC8253 ?

(d) Give feature of Alpha AXP processor.

(e) Give Alternate Functions of Port2 and Port3 of IC8051.

2. (a) What are different data transfer Schemes ?

Explain DMA data Transfer ?

(b) Describe different I/O addressing schemes available in 8085.

(c) Write a program for 8085 with proper comments to exchange memory block of 10bytes between 4000H onwards to 5000H onwards.

3. (a) Discuss the Internal architecture of 8086.

(b) Describe different Interrupts of 8086. Explain how Interrupt Vector Table is used ?

(c) Write a program for 8086 to multiply two 16 bit numbers and store the result in consecutive memory locations.

4. (a) Compare RISC and CISC architecture.

(b) Describe the protected mode of operation of IC80286.

(c) Write a short note on Super Scalar Architecture.

5. (a) Discuss the operation of IC 8255 in different modes with the help of its control word. Explain how and where BSR mode is used ?
- (b) Draw the register architecture of IC8257 and explain its operation.
- (c) Discuss the Interfacing of IC 8259 with the microprocessor and five its architectural features.
- When do we use it in cascade mode ?

6. (a) Describe how External RAM and ROM is interfaced with microcontroller IC8051.

(b) Explain different modes of operation of Timers in 8051 with related special function registers and start/stop mechanism of timers.

(c) Discuss the different addressing modes of IC8051.

